

System dynamics model for sustainable energy transition in the Republic of Benin

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ABSTRACT

Modern energy plays a vital role in national and global development, particularly by supporting industrialization and poverty reduction. However, planning for sustainable energy systems remains complex due to the interaction of various sectors, socio-economic factors, and energy resources. This study uses a systems dynamics approach to explore sustainable energy pathways in the residential sector of the Benin Republic. It employs causal relationships and feedback loops to analyze interactions among population, GDP, electricity demand and supply, CO₂ emissions, and techno-economic factors. The model spans 50 years (2013–2063), using 2013 as the base year, and was calibrated and validated with data from 2013 to 2023. The sensitivity analyses and scenario simulations were based on Benin's energy strategy and key national assumptions. The model identified leverage points to assess the impact of policies on electricity demand and supply. The results show that household electricity demand will increase significantly due to rising population and economic growth, reaching approximately 5249.85 MW in 2030 and 24,991.6 MW in 2063. However, energy efficiency improvements, renewable energy integration, and reduced transmission losses can mitigate this growth. Combining a 30% efficiency gain and a 45.40% renewable energy share could reduce the supply-demand gap to 698 MW by 2030. If transmission losses are reduced to 10%, the gap could shrink to just 51 MW by 2063. The findings emphasize the value of a systemic, dynamic approach to energy planning. The Stock and Flow Diagram model developed in this study serves as a decision-support tool to help policymakers design integrated strategies for sustainable electricity access and to support Benin's long-term energy transition goals aligned with Agenda 2063.

Introduction

Background

Energy is critical for all developmental processes and essential for almost all economic activities, improving living standards, enabling the delivery of essential services such as health care, education, and communication (Ouedraogo, 2013). However, many developing countries face significant challenges in achieving a reliable and sustainable energy supply. The Republic of Benin exemplifies this challenge, as its

energy system is characterized by limited domestic generation capacity, a heavy dependence on imported electricity, and low electrification rates, despite growing energy demand driven by population growth and economic development (Ligan et al., 2026).

Despite ongoing reforms and investments in the energy sector, electricity access in Benin remains relatively low and unevenly distributed across the country. For instance, the National electricity access is estimated at around 40%, with urban areas having significantly higher access at about 70% compare to the rural areas, which is extremely low (Akpahou et al., 2023; SEforALL, 2026). This disparity highlights the

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persistent energy poverty experienced by a large segment of the population and underscores the need for inclusive energy planning. Another major structural challenge in Benin's energy sector is the country's strong dependence on imported energy and electricity from neighbouring countries, like Nigeria, Ghana and Côte d'Ivoire. Estimates indicate that nearly 95% of the electricity consumed in Benin has historically been imported, exposing the country to energy insecurity (Mensah et al., 2022; World Bank, 2019).

To tackle these challenges, the Government of Benin has initiated several economic and energy reforms, including strategic initiatives aimed at improving electricity access, strengthening domestic energy production and decarbonizing its energy system by transitioning toward clean energy with plans to install up to 150 MW of solar energy by 2026 and to achieve universal electricity access by 2030 (Knowhow, 2025; Paulavets & Toure, 2026). National energy policies emphasize diversification of the energy mix, promotion of renewable energy technologies, and expansion of off-grid and mini-grid systems for rural electrification. Recent initiatives include the development of several solar power plants, rural solar mini-grids, and off-grid electrification programs supported by international partners such as the World Bank, the Millennium Challenge Corporation and the European Investment Bank.

However, achieving a sustainable energy transition in Benin requires addressing multiple interrelated factors, including energy infrastructure development, renewable energy investment, policy frameworks, technological innovation and socio-economic conditions (Bajja et al., 2025). These complexities justify the application of integrated modelling tools such as system dynamics. System dynamics allows researchers to represent feedback loops, delays, nonlinear relationships, and policy interventions within integrated simulation models (Forliano et al., 2024; Forrester, 1992). By capturing the interactions between energy and techno-economic variables, system dynamics models can help provide relevant sustainable energy insights to energy planners and policy-makers (Alemayehu et al., 2025).

Therefore, this study seeks to develop a system dynamics model to investigate sustainable energy transition in the Republic of Benin. The model will provide a comprehensive framework for analyzing the long-term dynamics of the country's energy system and identifying policy strategies capable of achieving reliable, affordable, and sustainable energy access. Ultimately, the research aims to support evidence-based decision-making that will accelerate Benin's progress toward universal electrification, renewable energy adoption and sustainable development.

Literature review

The transition from fossil-fuel-based energy systems to sustainable, renewable energy sources has become a central global development priority. This transition is driven by concerns over climate change, energy security, environmental sustainability and the need to achieve the United Nations Sustainable Development Goal 7 (SDG7), which seeks to ensure affordable, reliable, sustainable, and modern energy for all (UN-SDGs, 2018). Energy has found significant applications, and as such, presents concerns, and the outcomes pose significant complexities as they interact and create feedback systems that might be difficult to provide reliable policy insights on energy transition.

The study from Suganthi and Samuel (2012) identified energy as the most important environmental issue, ahead of water, food, and natural resources. A robust energy strategy and a reliable energy supply are essential for sustaining population growth, economic development, and energy demand. As a result, the government should ensure continuous access to energy to meet demand (Fotsing, 2014). Meeting energy demand requires sound energy planning, management, and forecasting efforts, as well as a thorough understanding of the factors that influence energy use (Li et al., 2011). Several approaches and methodologies have been employed to investigate the energy planning, for instance, Atchike et al. (2020) investigated the relationship between electricity

consumption and foreign direct investment and economic growth in Benin using an autoregressive distributed lag (ARDL), bounds test with dummy variables, and the Toda-Yamamoto approach to show evidence of unidirectional causalities from electricity consumption to both economic development and foreign direct investment. Marcel (2019) suggested that the government of the Republic of Benin should target the implementation of new strategies to improve access to electricity to attract more foreign investment and achieve rapid, sustainable economic growth.

Wolde-Rufael (2006) investigated a bidirectional causality from electricity consumption to economic growth and showed that energy consumption is an indispensable component of economic growth, a direct and indirect input in the production process, and a complement to capital and labour. Al-mulali and Binti Che Sab (2012) investigated the role of Carbon dioxide (CO₂) on energy consumption, which is associated with economic growth. Ouedraogo (2013) explored the bidirectional causality between energy consumption and GDP in Benin, while Nkengfack and Fotio (2019) established a long-term relationship between energy consumption and economic growth. This correlation suggests a close relationship between economic development and energy.

In 2000, Benin announced its challenges for the horizon 2025 referred to as 'Benin 2025 Alafia'; it aims at achieving a prosperous, competitive, well-governed and peacefully united nation by 2025 (Tchané, 2018). The 'Benin Vision 2025 Alafia' is an all-inclusive, comprehensive national development plan that establishes mid-term goals across various sectors, such as socio-economic, political, energy, etc. Also, the Government Action Program (PAG), which ran for five years between 2021 and 2026, is based on the strategic objectives of the BNDP between 2018 and 2025 and aims to accelerate Benin's economic and social development. The PAG is the only operational planning instrument at the national level. As such, it is the sole instrument for steering government action or guiding the activities of Ministries, Departments, Agencies and other parastatals, as well as developing and implementing the state budget over the period (Programme d'Action du Gouvernement 2021-2026, 2021).

In all the national policy documents and action plans, national economic growth is expected in the development of agribusiness, services and tourism within a framework of more effective national and local governance, relying on the development of human capital and infrastructure (Ikeije & Okechukwu, 2025). To achieve these interrelated policies, there is a need to consider the dynamic nature of national policies that are susceptible to change and resistance over time. As such, there is a need to consider a systemic planning approach that integrates the main challenges and issues of the Sustainable Development Goals (SDGs) and the African Union Agenda (AU) 2063, especially in the energy sector, because the sector is a major driver to achieving most of the SDGs and the AU Agenda 2063 (Chimhowu, 2021). A very significant aspect is having a holistic estimation of energy demand and supply to create an effective energy policy and support utility management. Achieving a holistic energy management system is crucial because a balanced electricity demand-supply will ensure the security of the energy sector (Amusa et al., 2009). Also, Akpahou et al. (2024) used a linear modelling approach in the EnergyPLAN framework to investigate sustainable energy planning in West Africa, specifically analyzing long-term energy demand and renewable energy (RE) deployment in Benin. However, the challenges in energy policy and energy security are not always linear, as a few authors claim (Guenoupkati et al., 2021; Sotiman Yotto et al., 2021). Also, evidence from most of the studies reviewed and presented in the literature has used linear regression approaches and predominantly used energy demand forecasting methods (e.g. EnergyPLAN model) (Akpahou et al., 2024). However, Laimon et al. (2022) claimed that linear approaches may present a myopic modelling paradigm that will not comprehend the complexity and underlying rationale of energy systems.

The policy challenges often encountered in the power sector are

uniquely dynamic because the sector operates as a complex, inter-connected system, solving one problem frequently introduces new, often unforeseen, technical, economic, or social challenges (Somoye, 2023). Thus, many policy solutions create unintended problems, such as policy resistance and delays (Sterman, 2000). As such, numerical models have failed to identify this caveat in energy planning and policy; hence, the need for a more holistic, systems-based approach to planning energy systems and providing sustainable approaches in the energy sector.

Understanding the complex interactions between energy demand, technological development, policy interventions and socio-economic factors requires a systems-based modelling approach, such as system dynamics. Several studies have employed a systems dynamics (SD) model to investigate sustainable energy frameworks, as SD modelling is a widely used, evidence-based approach for analyzing sustainable energy systems, managing energy transitions, and evaluating policy impacts. For instance, Alemayehu et al. (2025) used the SD framework to formulate energy policies for sustainable development in Ethiopia and according to Ahmad et al. (2016), the SD modelling framework has also been widely applied in the electricity sector in many developing countries' case studies to study transition in clean cooking (Shari et al., 2022), electricity sector (Moumouni et al., 2014; Shari et al., 2018). Also, it has been considered to model the West Africa energy transition framework (Momodu et al., 2017), including investigating low-carbon transition in the residential sector (Shari et al., 2020).

Despite increasing research on renewable energy development in Benin, several gaps remain in the literature, as most studies focus on technology-specific assessments rather than integrated national energy systems. Also, existing research primarily relies on optimization models that do not adequately capture feedback mechanisms within energy systems, with limited integration of socio-economic variables.

This study seeks to address these gaps by developing an integrated system dynamics model that captures the complex interactions between energy demand, renewable energy supply, technological development, policy interventions and environmental impacts. Hence, the study will contribute to existing knowledge in energy planning of a typical low-income countries in sub-Saharan Africa, in the case of Benin by: (i) investigate the sustainable energy agenda of the Republic of Benin, (ii) identifies the main factors that affect the energy demand of Benin, and (iii) explores their interconnection and socioeconomic impact on the country's energy demand and sustainable energy mix.

Methodology

The methodology section presents the SD framework of the Benin energy sector, with a focus on household energy demand. The SD model was developed using the Vensim PLE software.

Model development of the systems dynamics

Modelling framework

After thoroughly examining Benin's energy system and clarifying the intended purpose of the modelling framework. The model boundary was defined by identifying the key techno-economic components required to capture the dynamic behaviour of the Benin energy system. The model framework and boundary focus solely on modern energy (electricity) and include three generation technologies available in Benin: small/medium hydropower, natural gas turbines and solar photovoltaics (PV). It also considers electricity imports and transmission losses. Net migration is excluded from the model boundary due to its negligible impact on the total population and the lack of reliable annual migration data. The study covers a 50-year modelling framework, i.e., 2013 to 2063. It is important to note that 2013 was chosen as the modelling base year because there is stable, relatively complete energy data available for that year. Fig. 1 shows a typical system dynamic modelling framework, represented with five thematic activities. The study commenced by articulating the problem, ensuring that the problem to be addressed

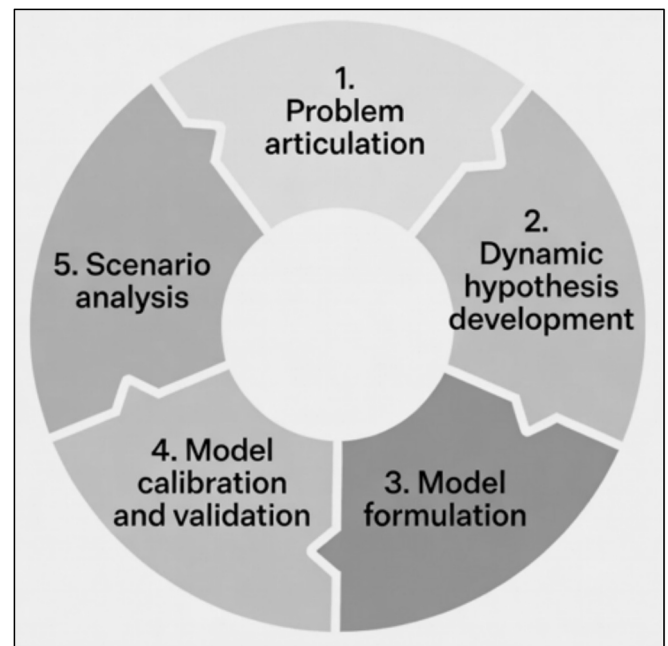


Fig. 1. The current study framework flowchart. (Adopted from Sterman (2000).)

was put in the right context and perspective. The second step is developing a dynamic hypothesis, which implies that the study provided an empirical assumption to be tested or investigated through the development of a causal loop diagram. Step three is where the model, i.e., the Stock and Flow diagram, is developed or designed. Calibration and validation of the model are conducted in the fourth step. The final step is the scenario analysis, in which the developed model is tested using identified leverage points to provide relevant policy insights.

System variables

Secondary data was primarily used for the analysis. Key variables, also referred to as leverage variables, are those capable of triggering significant changes across the system, though often difficult to identify in complex systems (Amblard et al., 2015). They are essential for understanding model behaviour and validating its structure and boundary adequacy. Hence, a comprehensive review of relevant literature, national reports, and scientific studies was conducted, supplemented by multiple regression analysis to identify socio-economic and technical variables. Secondary data were gathered from both published and unpublished sources aligned with the data requirements of the study. Major sources include the report from the Energy Information System on Energy Balances and Performance Indicators (1990–2023) from the General Directorate of Energy Planning and Rural Electrification (Sie, 2024). The national master plan for developing the electricity sub-sector in Benin, and reports from the National Institute of Statistics and Demography (IED, 2015). Variable names, notations, and notations used in the Stock and Flow Diagram (SFD) are summarized in Table 1. The differential equations of the simulation were developed based on actual baseline year data, enabling dynamic modelling of household electricity demand and efficiency in Benin.

Causal Loop Diagram (CLD)

Understanding the behaviour of a system requires a clear comprehension of its elements and the interactions between them. A powerful method for visually representing the feedback relationships among system components is the causal loop diagram (CLD). The primary purpose of a CLD is to illustrate how certain elements influence others within a dynamic system (Sapiri et al., 2017). These causal loops can be

Table 1
Mathematical notations and nomenclatures of SD-SFD.

Variable	Unit	Type	Notation
Household Electricity Demand	kW	Stock	HHED
Benin Population	People	Stock	BP
GDP	FCFA/person	Stock	GDP
HH Electricity supply	kW	Stock	HHES
Changes in Birth Rate	People/year	Flow	CBR
Changes in the Death rate	People/year	Flow	CDR
GDP growth	FCFA/year	Flow	GG
Changes in Household electricity demand	MW/year	Flow	CHED
Changes in supply	MW/year	Flow	CS
Changes in consumption	MW/year	Flow	CC
Average wage bill	Per year	Rate	AWB
Fractional birth rate	Per year	Rate	FBR
Fractional death rate	Per year	Rate	FDR
GDP growth rate	Per year	Rate	GDPGR
Rate of Household electricity demand change	MW/year	Rate	RHHED
Rate of electricity supply		Rate	RES
Consumption rate	Per year	Rate	CR
Transmission losses	Per year	Rate	TL
CO ₂ emission	kt Eq CO ₂	Auxiliary	
Residential Electrified Population	People	Auxiliary	REP
Per capita Electricity consumption	kWh	Auxiliary	PEC
Investment in generation	F CFA/year	Auxiliary	IG
Demand-Supply gap	kW	Auxiliary	DSG
Electricity Imports	kW	Auxiliary	EI
Small/medium hydroelectricity production	kW	Auxiliary	SMHP
Solar PV production	kW	Auxiliary	SPV
Thermal power plant	kW	Auxiliary	TPP
Unelectrified Residential Population	People	Auxiliary	URP
National income per capita	FCFA/person	Auxiliary	NIPC
CO ₂ emission factor	kt Eq CO ₂ /MW	Auxiliary	EF
National income	FCFA	Auxiliary	NI
Generation	kW/year	Auxiliary	G

categorized as either balancing (self-correcting or negative feedback) or reinforcing (positive feedback) (Bayer, 2004). A balancing loop (negative feedback) occurs when a change in one variable is counteracted by the loop's feedback effect, thereby promoting system stability (Bayer, 2004). In contrast, a reinforcing loop (positive feedback) amplifies a change in one variable through feedback, accelerating system behaviour in the same direction (Bayer, 2004).

In this study, the CLD was developed in four key stages. First, we identified the main issues of the energy sector of Benin (referred to as variables) through multiple regression analysis, literature reviews, media content, and policy documents. In the second stage, these variables were linked to form an initial CLD, showing their interconnections and polarities. The third stage involved reviewing and refining the diagram through consultations with several experts from the Beninese energy sector, during which the preliminary CLD was broken into specific feedback loops and revised based on expert feedback regarding variable relationships and linkages. Finally, this working CLD was further refined to produce the final version, representing the dynamics of the residential electricity sector of Benin.

Stock and Flow Diagram (SFD)

It is important to emphasize that the structure of a system fundamentally shapes its dynamic behaviour. A Stock and Flow Diagram (SFD) offer a more detailed and explicit structural depiction of a system than a Causal Loop Diagram (CLD). Stocks represent accumulated quantities that define the system's state at any given moment. They are initialized with specific values and can only evolve through associated inflows and outflows. Stocks introduce delays and inertia into the system, serving as key elements for generating the information required for decision-making and action (Bayer, 2004). In contrast, flows determine the rate at which stocks change. These flows indicate how quantities accumulate or deplete over time and are measured in terms of change

per unit of time (Sapiri et al., 2017).

The model is organized according to the structure reflected by the CLD. With the Vensim software, the causal relationships identified in the CLD are translated into a more detailed SFD diagram, integrating stocks and flows corresponding to variables in Benin's residential sector. Stocks include variables such as household electricity consumption and installed production capacity, which accumulate or decrease depending on changes in supply and demand. Flows represent elements such as the population growth rate, investments in energy production, or variations in energy consumption due to economic changes.

Model testing and validation

After the model was developed, it underwent a validation process to assess its reliability. Validation in SDs focuses on evaluating how the model could most likely replicate the expected behaviour of the real system, not necessarily achieving a perfect match with reality (Sapiri et al., 2017; Zita, 2021). In this study, the dynamic model was validated over the period 2013 to 2023 by comparing simulated outputs with historical data for per capita electricity consumption from the simulated data and the official/observed/historical energy data. As shown in Fig. 2, the simulation results were plotted alongside observed values; both historical and simulated data showed a consistent positive trend. Correlation reached 0.92206 for historical data and 0.99609 for simulated output. There is a relative error margin of about 8.03%, which is within a permissible range for an exploratory study. These high correlation values indicate that the model closely mirrors real-world trends, despite minor discrepancies, as observed in Sapiri et al. (2017). The key aim of validation is to ensure the model behaves in a way consistent with expectations. In this work, the validated model effectively reproduces the system's dynamics and is reliable for forecasting future scenarios.

Results, discussions, scenario development and sensitivity analysis

Socio-economic variables and sustainable energy pathways through a systems thinking paradigm

The Causal Loop Diagram (CLD) presented in Fig. 3 shows the structural explanations of the interconnection of the variables, revealing how model variables interact closely with real-life situations and how they communicate with the dominant feedback loops within the model. In the subsection, the CLD was used to explain the interactions and relationships among the socioeconomic and environmental variables and their dynamic components, and to illustrate how they can be used to assess the sustainability of the household power sector. This approach is referred to as systems thinking in SDs.

The Causal Loop Diagram

The CLD, a structural illustration of the system thinking paradigm, is shown in Fig. 3. This enables us to achieve the first objective of the study, which is to reveal relevant social and economic variables and their interaction to enhance a sustainable household energy pathway in Benin. The CLD is further discussed from the perspective of Reinforcing and Balancing loops. It is important to note that the CLD is a qualitative assessment of the dynamic system; it will only provide clarity to relevant variables and enable the identification of necessary leverage points. Thus, leverage variables are variables to which policy actions could be applied. Any changes in these variables will present significant policy implications in the system.

Reinforcing loop. A reinforcing loop (R-loop) in a CLD is a feedback loop that amplifies changes in a system. When a variable increases, it triggers a series of effects that ultimately reinforce the initial change, leading to growth or decline over time (Bayer, 2004). The R-loops identified in the CLD are:

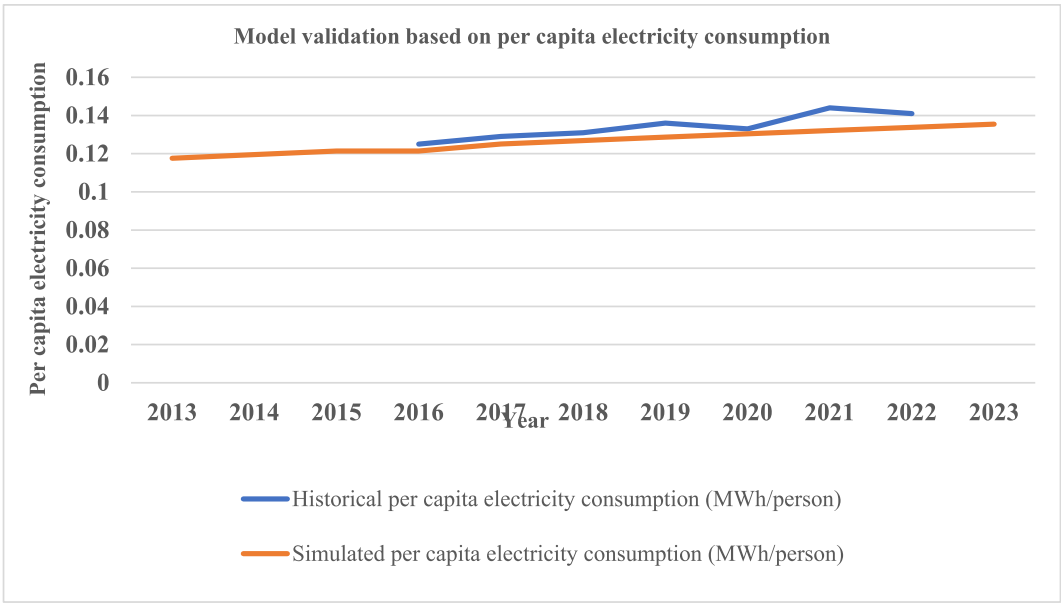


Fig. 2. Simulated and historical per capita electricity consumption of Benin.

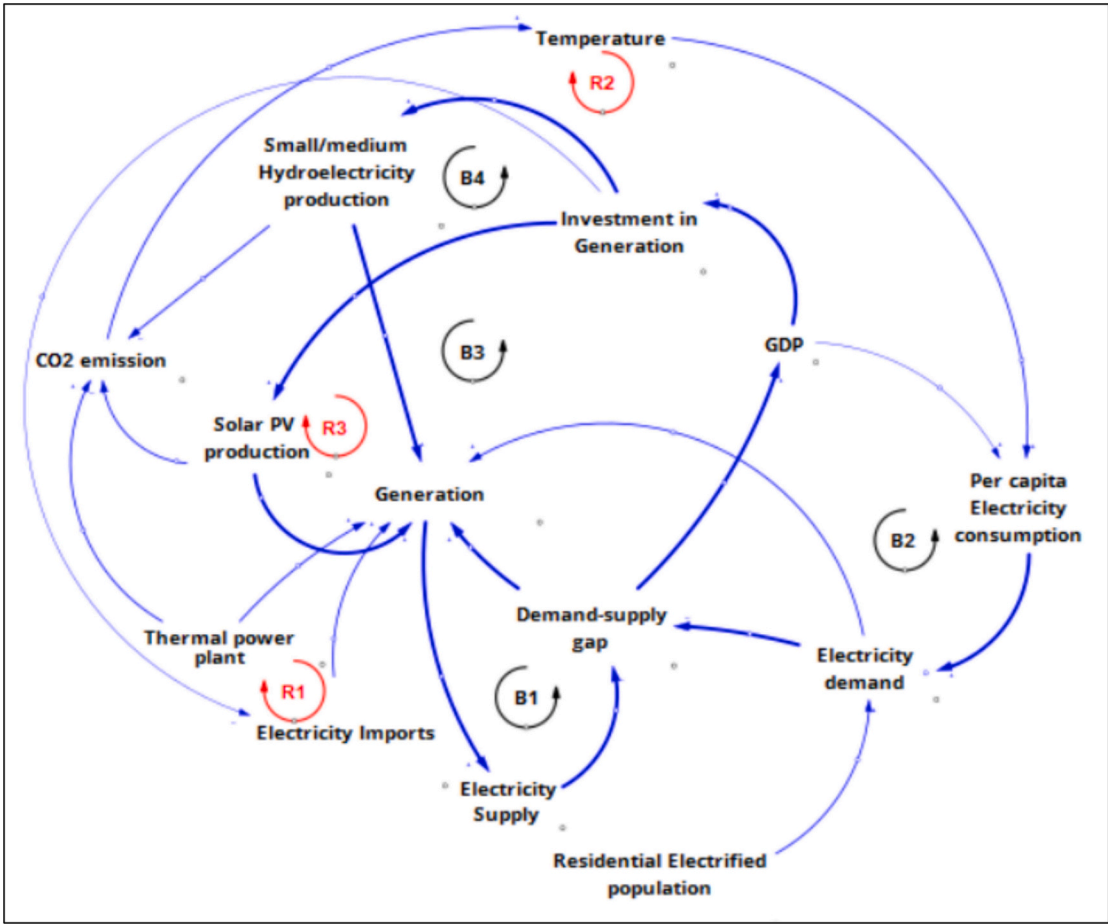


Fig. 3. Causal Loop Diagram of the model.

R1: Electricity Supply, Demand-supply gap, GDP, Investment in Generation, Electricity Imports, Generation, and Generation.

R2: Electricity Supply, Demand-supply gap, GDP, Investment in Generation, Small/medium Hydroelectricity production, CO₂ emission,

Temperature, Per capita Electricity consumption, Electricity Demand, Generation, Electricity Supply.

R3: Electricity Supply, Demand-supply gap, GDP, Investment in Generation, Solar PV production, CO₂ emission, Temperature, Per capita

Electricity consumption, Electricity Demand, Generation, and Electricity Supply.

The links and interactions were identified thus: Electricity Supply → Demand-Supply Gap → GDP → Investment in Generation → Small/Medium Hydroelectricity Production → CO₂ Emission → Temperature → Per Capita Electricity Consumption → Electricity Demand → Generation → Electricity Supply, represents the reinforcing loop (R2), which reveals potential policy action for the associated electricity demand and environmental factors. The loop creates a self-sustaining pattern of energy consumption growth. For instance, the loop R2 aims to meet the electricity gap by ensuring local energy security. Also, investing in electricity generation or importing electricity could be an alternative way to address energy shortages when the electricity supply struggles to meet demand and the demand-supply gap widens.

However, as energy production from fossil fuels increases, it contributes to higher CO₂ emissions, which in turn lead to rising temperatures. Higher temperatures drive up per capita electricity consumption, particularly due to the greater need for household cooling. This increase in electricity demand constrains generation capacity, ultimately exacerbating the demand-supply gap. The loop continues to reinforce the system as economic growth and rising temperatures perpetuate an ever-growing energy demand, requiring continuous investment in generation capacity.

Balancing loop. A balancing loop (B-loop) in a CLD represents a feedback process that stabilizes or regulates a system. When a variable increases, the loop generates a counteracting effect that slows down or limits further change, preventing uncontrolled growth or decline (Bayer, 2004). The balancing loops identified in the CLD are:

B1: Electricity Supply, Demand-supply gap, Generation, and Electricity Supply.

B2: Electricity Supply, Demand-supply gap, GDP, Per capita Electricity consumption, Electricity demand, Generation, and Electricity Supply.

B3: Electricity Supply, Demand-supply gap, GDP, Investment in Generation, Solar PV production, Generation, and Electricity Supply.

B4: Electricity Supply, Demand-supply gap, GDP, Investment in Generation, Small/medium Hydroelectricity production, Generation, and Electricity Supply.

The balancing loop (B3), on the other hand, introduces a stabilizing mechanism by ensuring that investments in renewable energy contribute to closing the demand-supply gap. As electricity supply falls short of demand, the gap widens, affecting economic performance and triggering a decrease in GDP. In response to rising financial demands, policymakers and stakeholders are allocating more investment to energy generation, with a particular focus on solar PV as a renewable alternative. The expansion of solar PV production increases electricity generation capacity, gradually reducing the demand-supply gap. By increasing the availability of electricity through sustainable means, this loop mitigates the supply-demand imbalance and ensures a reliable energy system.

Results and discussion

The Benin household energy demand pattern, investigated using the CLD's system thinking paradigm, is modelled across different scenarios using an SD modelling framework. The SD model employed in this regard is the Stock and Flow modelling approach. The application of the SFD, therefore, is presented in this section.

Stock and Flow Diagram (SFD)

The dynamic model of the Benin residential electricity sector in this work is developed through an SFD, as shown in Fig. 4. The dynamic model is used to analyze the quantitative outcome of the Benin household energy demand. The SFD model is composed of level or stock variables (Household Electricity demand, Benin Population, GDP, HH Electricity supply), rate variables (inflows: Average wage bill, Fractional birth rate, Changes in Birth rate, Fractional death rate, Changes in Death rate, GDP growth rate, Changes in Household electricity demand, Rate of Household electricity demand change, Changes in supply, Rate of electricity supply, Changes in consumption, Consumption rate, Transmission losses), and auxiliary variables (CO₂ emission,

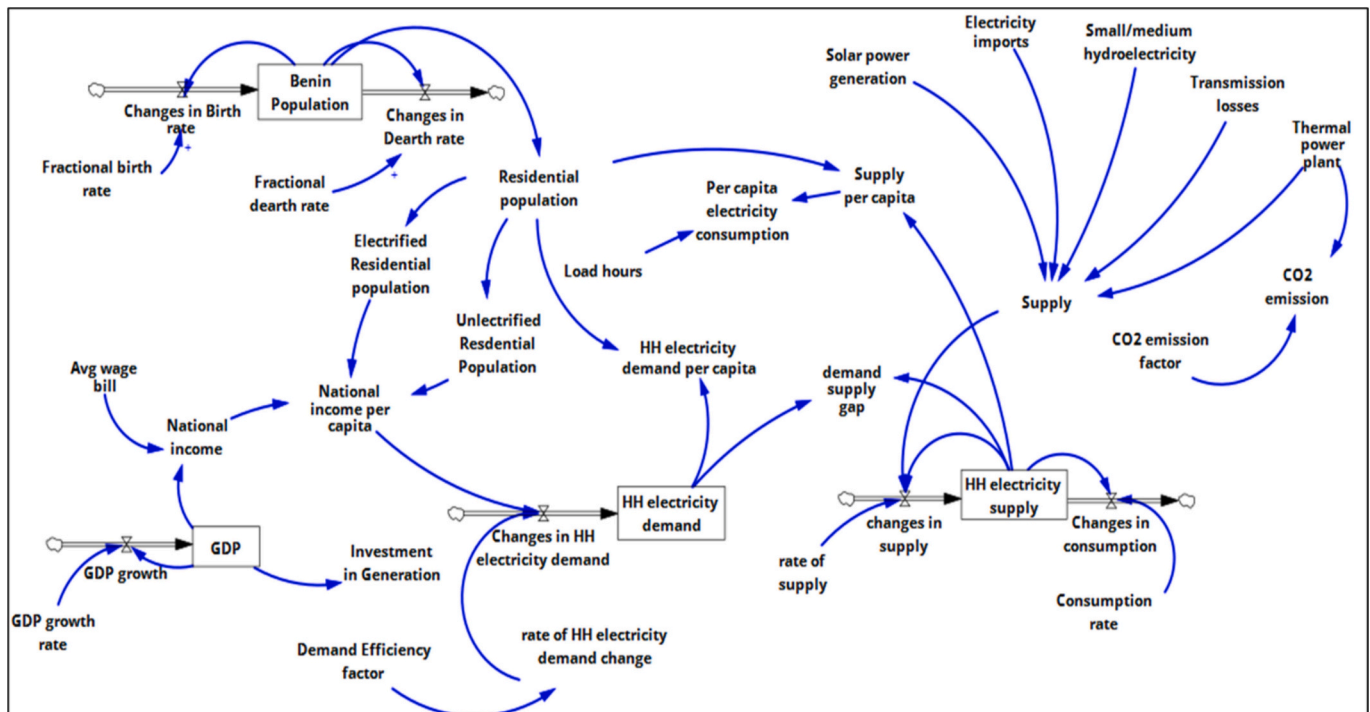


Fig. 4. Stock and Flow Diagram (SFD) for the Benin household energy system.

Residential Electrified Population, Per capita Electricity consumption, Investment in generation, Demand-Supply gap, Electricity Imports, Small/medium hydroelectricity production, Solar PV production, Thermal power plant, Unelectrified Residential Population, National income per capita, CO₂ emission factor, National income), all with a year time step.

The model incorporates the dynamics of electricity supply, demand, population, and GDP to analyze sustainable energy pathways. The electricity supply represents the stock of energy/power generated from both conventional and renewable energy sources, such as thermal power plants, solar PV, hydroelectricity, and imports, while electricity demand is influenced by economic activity (GDP) and population growth. Investment in energy infrastructure is crucial to expanding electricity generation capacity. The demand-supply gap influences investment decisions, thereby altering the rate of electricity generation. The model also accounts for environmental factors, such as CO₂ emissions from thermal power plants and temperature variations, which affect electricity demand (e.g., cooling needs).

Fig. 3 illustrates the dynamic relationship between GDP and energy demand. Economic growth is linked to energy consumption (with a significant impact), as higher energy use typically supports greater economic output. As national GDP rises due to expanded economic activity, GDP per capita also increases. This growth in economic activity leads to higher energy consumption per capita, which in turn drives up electricity demand.

Furthermore, the population stock grows through births and declines through outflows (deaths). As the population grows, demand for electricity rises, requiring greater generation capacity. Higher per capita electricity consumption further amplifies household demand, underscoring the need for an adequate electricity supply to ensure long-term energy security. As household electricity demand increases, there is a corresponding need for increased supply; however, caution should be taken not to rely solely on thermal or mono-supply energy sources, or to be over-dependent on electricity imports from neighbouring countries. This suggests that the country needs to consider other energy sources, especially renewable sources such as wind and hydropower (medium- and small-scale). Consideration of alternative energy sources will enable a reliable energy mix and ensure the security of supply in the long run. The SFD has therefore considered current electricity sources and proposed the impact of future implications of renewable energy sources on the overall energy security and sustainability of the system.

Household energy demand pattern

This subsection presents results and discusses the analysis of the Benin household energy demand pattern based on the aforementioned dynamic variables, including GDP, population, and efficient energy use.

Population and household electricity demand. A sensitivity analysis of the population was conducted under the assumption that population growth would increase. The population growth percentages used in the sensitivity analysis are shown in Table 2.

Fig. 5 illustrates the impact of population growth on household electricity demand in Benin. The findings indicate a positive correlation between the two variables: as the population increases, household electricity demand also rises. Under the business-as-usual scenario,

Table 2
Population growth rate at different growth rates.

SN	Population growth	Fractional birth rate	Annotation
1	0%	0.0364	Business as usual (BAU)
2	10%	0.04004	Low Growth Rate (LGR)
3	15%	0.04186	Low-Medium Growth Rate (LMGR)
4	20%	0.04368	Medium Growth rate (MGR)
5	30%	0.04732	High Growth Rate (HGR)

demand is projected to reach approximately 5249.85 MW by 2030, and 24,991.6 MW by 2063. This relationship is explained by the fact that a growing population requires more energy to meet daily needs, resulting in a proportional rise in electricity demand.

GDP and electricity demand. The sensitivity analysis of GDP was conducted under the assumption that the percentage of population growth in GDP was increasing. The percentage of GDP growth used in the sensitivity analysis is in Table 3.

Fig. 6 illustrates the impact of GDP growth on household electricity demand in Benin. The analysis reveals a positive relationship: as GDP increases, household electricity demand also rises. In other words, economic growth directly contributes to higher electricity consumption. This increase can be seen in trends of HH electricity demand at a Low GDP Growth Rate (LGR), Low-Medium GDP Growth Rate (LMGR), Medium GDP Growth Rate (MGR), and High GDP Growth Rate (HGR) compared to the base, representing the HH electricity demand at a business-as-usual scenario where there is no increase in the GDP growth rate. Energy plays a crucial role in enabling countries to sustain economic growth, and as the economy expands, energy demand inevitably increases. Based on the results, Benin's GDP is projected to reach approximately 15.71 trillion by 2030.

Energy efficiency and HH energy demand. The Demand Efficiency Factor (DEF) is a coefficient that quantifies improvements in energy efficiency from electricity consumption and the use of efficient appliances. It represents a system or sector's ability to reduce energy consumption while maintaining the same level of service or production. The sensitivity analysis of electrical appliances was conducted with the assumption that energy efficiency was increasing. The factor ranges from 20% to 26% (as shown in Table 4).

Fig. 7 shows the sensitivity analysis of household electricity demand across five sensitivity runs, using the values presented in Table 4. This sensitivity analysis shows that increased energy efficiency in the Beninese household sector would lead to a significant reduction in electricity demand. By 2063, the projected reduction in electricity demand due to energy efficiency measures is estimated at 2804.4 MW under a low-growth rate, 4085.4 MW under a low-medium-growth rate, 5291.7 MW under a medium-growth rate, and 7497.8 MW under a high-growth rate scenario. This reduction in electricity demand could be reallocated to regions with low electrification rates, thereby improving overall electricity access.

This trend is evident in the variations in household electricity demand across different energy-efficiency growth scenarios. Specifically, electricity demand under low, low-medium, medium, and high-energy-efficiency growth rates exhibit a downward trend relative to the BAU scenario, which serves as the baseline with no energy-efficiency measures implemented.

Energy mix for sustainable energy security by 2030 and AU goals by 2063

The BAU shown in Fig. 8 reveals an increased reliance on imports to cover the energy deficit, as per the energy generation mix plan under the BAU scenario. Furthermore, thermal production is growing rapidly, yet 100% of the petroleum products, solid fuels and gas consumed in Benin are imported, which once again justifies the fact that this ambition is unrealistic from Sie (2024). However, such dependence makes the country vulnerable to fluctuations in costs and uncertainties in external supply. It is therefore essential to accelerate investments in renewable energy, particularly solar and hydropower, to increase national production and reduce energy dependence.

Increase in renewable energies in the energy mix-scenario. In this scenario, which is based on massive and sustained investment in renewable energies, we consider that these will account for approximately 35% of the national energy mix by 2030 (IED, 2015; MERPMEDER, 2015). Beyond

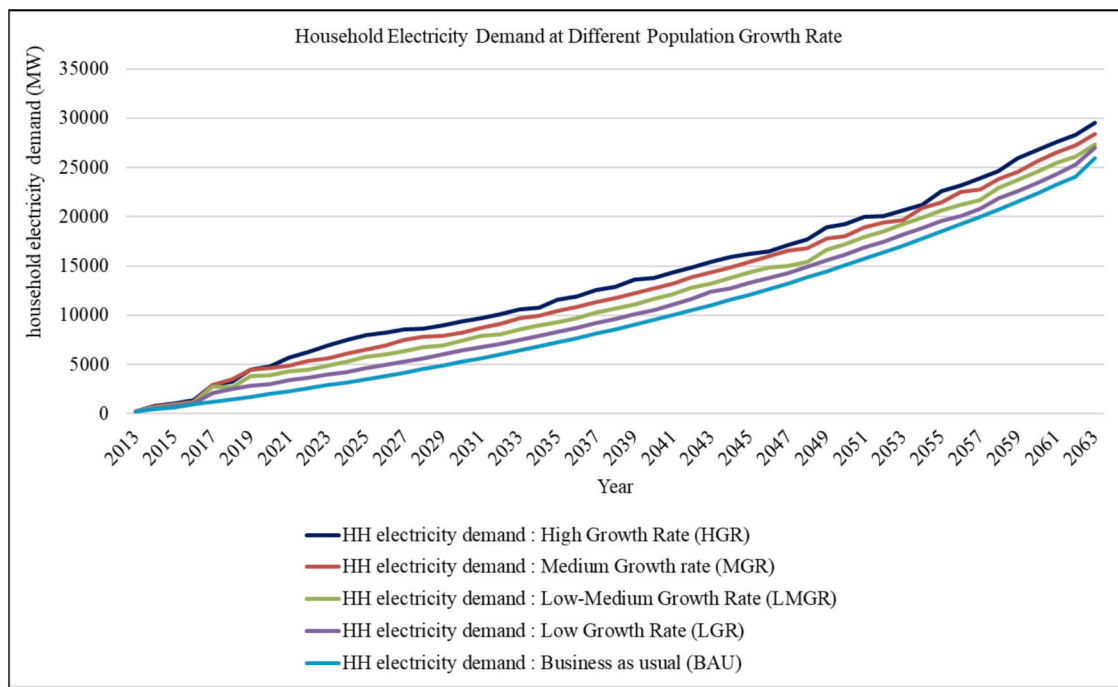


Fig. 5. Effects of population on household electricity demand.

Table 3

GDP growth rate at different growth paradigms.

S/ N	GDP growth	GDP growth rate	Annotation
1	0%	0.056	Business as usual (BAU)
2	10%	0.0616	Low GDP Growth Rate (LGR)
3	15%	0.0644	Low-Medium GDP Growth Rate (LMGR)
4	20%	0.0672	Medium GDP Growth rate (MGR)
5	30%	0.0728	High GDP Growth Rate (HGR)

2030, their share of the energy mix would continue to grow, reaching 100% by 2063. This transition will be accompanied by a gradual reduction in electricity imports, thereby strengthening the country's energy independence.

According to Fig. 9, there is a significant increase in household electricity supply with 2970 MW and 18,370 MW by 2030 and 2063, respectively, compared to the BAU scenario (1572 MW by 2030, and 4392 MW by 2063), while simultaneously reducing the country's dependence on external energy sources (such as imported electricity).

In this modelling, electricity imports are gradually reduced to a minimum level of 250 MW by 2063, while the share of renewable energy, particularly solar photovoltaic and hydropower, reaches 100% of the energy mix by 2063 (Fig. 10). According to Fig. 10, the installed capacity of solar photovoltaic systems will increase more rapidly than that of hydropower, and imports will continue to decline until they are near-eliminated in 2063. However, thermal power plants will remain dominant in the short and medium term, as they will continue to provide baseload generation necessary for the stability of the national electricity grid. The intermittency of solar PV and the 100% dependence on petroleum products, solid fuels, and gas consumed in Benin mean that imports are not eliminated from the energy mix but are reduced to the barest minimum.

Such a scenario will require appropriate energy policies, an incentive-based regulatory framework, and significant technological investments to ensure the efficient development of renewable infrastructure and the stability of the national electricity grid. Through these massive investments in renewable energy, Benin would significantly

reduce its energy dependence while substantially increasing its domestic production, thereby ensuring its long-term energy security.

In this scenario, the objective is to combine a 30% improvement in energy demand efficiency with an increase in investments in renewable energies, which would account for 45.40% of the energy system. This approach aims to optimize energy consumption while strengthening the share of renewable sources in the national energy mix.

The study found that after the joint implementation of measures to improve energy demand efficiency and increase the share of renewable energy to 45.40% in the electricity mix, the gap between electricity demand and supply is reduced to 698 MW in 2063, as shown in Fig. 11. This gap that persists between electricity demand and supply could be explained by the electric power transmission and distribution losses of 21.5% of output in 2019, which is higher than the average of 8.3% for sub-Saharan Africa, according to the World Bank (World Bank, 2020). This significant reduction in the energy deficit creates favourable conditions for universal access to electricity and improved energy security, and does so faster than if each measure were applied separately. To consolidate these advances and maximize their long-term effects, it is imperative to maintain these combined actions and gradually intensify investments in renewable energy infrastructure.

Effect of transmission losses on household (HH) electricity supply. Transmission losses (TL) refer to the electrical energy that is lost during its transmission through the electricity network, from generation plants to end consumers. In the sensitive analysis, the parameter data for the base scenario were kept constant at TL (0.215) in 2019, while varying those of the improved scenario, as shown in Table 5. These improved values come from Benin's National Energy Management Policy 2020–2030 (Ministère de l'énergie, 2020).

Fig. 12 revealed that as transmission losses decrease, the electricity supply increases. This implies that there is a negative relationship between transmission losses and electricity supply. The electricity supply at TLMDR (547.83 MW in 2030 and 2370.5 MW in 2063) and at TLDR (566.55 MW in 2030 and 2471.76 MW in 2063) is increasing compared to the electricity supply at TBAU (523.49 MW in 2030 and 2238.85 MW in 2063).

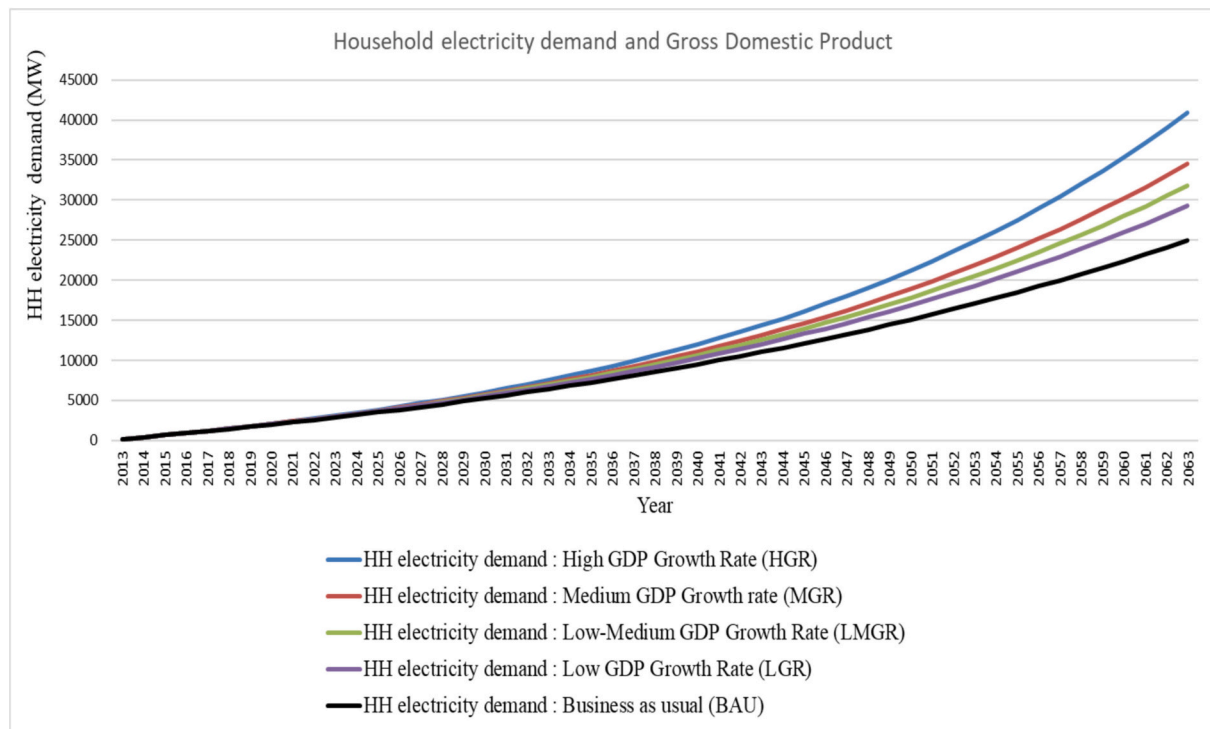


Fig. 6. Effects of GDP on household electricity demand.

Table 4
Sensitivity values on the energy efficiency factor and the energy demand.

SN	Efficiency factor growth	Efficiency factor	Annotation
1	0%	0.2	Energy Efficiency Business as Usual (EBAU)
2	10%	0.22	Energy Efficiency Low Growth Rate (ELGR)
3	15%	0.23	Energy Efficiency Low-Medium Growth Rate (ELMGR)
4	20%	0.24	Energy Efficiency Medium Growth rate (EMGR)
5	30%	0.26	Energy Efficiency High Growth Rate (EHGR)

Energy efficiency factor, transmission losses and demand-supply gap. To further optimize the electrical grid, we considered the energy efficiency factor and demand-supply gap scenario, where an additional improvement was achieved by reducing transmission losses to 10%. This led to the development of a new scenario: Energy Efficiency Factor, Reduced Transmission Losses, and Demand-Supply Gap. The objective of this scenario is to assess the impact of transmission efficiency on closing the demand-supply gap and improving electricity availability. Fig. 13 confirms that the share of renewable energy in the energy mix and transmission loss reduction could have a significant effect on the electricity demand-supply gap. Similarly, the results in Fig. 13 also indicated a significant reduction in the demand-supply gap (51 MW by 2063) compared to the previous Energy Efficiency Factor and Demand-Supply Gap scenario, which showed a 698 MW gap in 2063. Improvement in energy efficiency reduce demand-supply gap, thus improved transmission efficiency plays a crucial role in enhancing electricity availability (de la Rue du Can et al., 2022; Kelechi et al., 2025).

Discussion

The current study has provided a reliable policy intervention framework that focuses on enhancing demand-side efficiency, reducing

transmission and distribution losses, and increasing investment in renewable energy sources, thereby improving Benin's overall energy security. Moreover, the study has revealed that the systems thinking and system dynamics (SD) approach is effective in understanding Benin's energy system and ensuring the sustainability of the household power sector. Ohanu et al. (2024) supported this claim that a systemic approach of integrating infrastructure improvements, energy diversification and technological innovations is sacrosanct to achieve a sustainable and resilient electricity supply for all households. More so, an elaborate SD study by Varma and Sushil (2019) and another from Ale-mayehu et al. (2025) revealed that SD is a reliable methodology and has presented (i) a robust electricity demand-supply model for policy making, (ii) a sufficiently insightful projection of electricity demand driven by GDP, population and resource; (iii) ability of handling multiple non-linear causal relationships, delays and feedbacks, and (iv) and ensure Energy security & environmental concerns answered with the system dynamics model. Thus, claims from Varma and Sushil (2019) and Ale-mayehu et al. (2025) further validate the current study's modelling framework – SD. In addition, De Rosa et al. (2024) support the study's system-thinking approach, in which relevant variables were integrated to investigate the causal relationships among several factors affecting sustainable energy security. Hence, De Rosa et al. (2024) revealed that energy supply and demand, as well as human capital accumulation, can impede sustainable development. Also, Adams et al. (2020) support the claim from the study that transmission and distribution decrease economic growth, and Adams et al. (2020) revealed that losses in transmission and distribution decrease economic growth by about 40%. Similarly, Osei-Gyebi and Dramani (2023) supported that electricity transmission losses had a significant effect on electricity consumption and economic growth in Ghana. Further, Shari et al. (2020) found that understanding the relationships among residential electricity consumption, energy efficiency, and carbon footprint tends to promote energy-saving behaviours and provide a clear distinction between electricity consumption and demand, which slightly undermines claims in the current study. This contradiction could result from the use of electricity tariffs, which create a dichotomy between consumption and

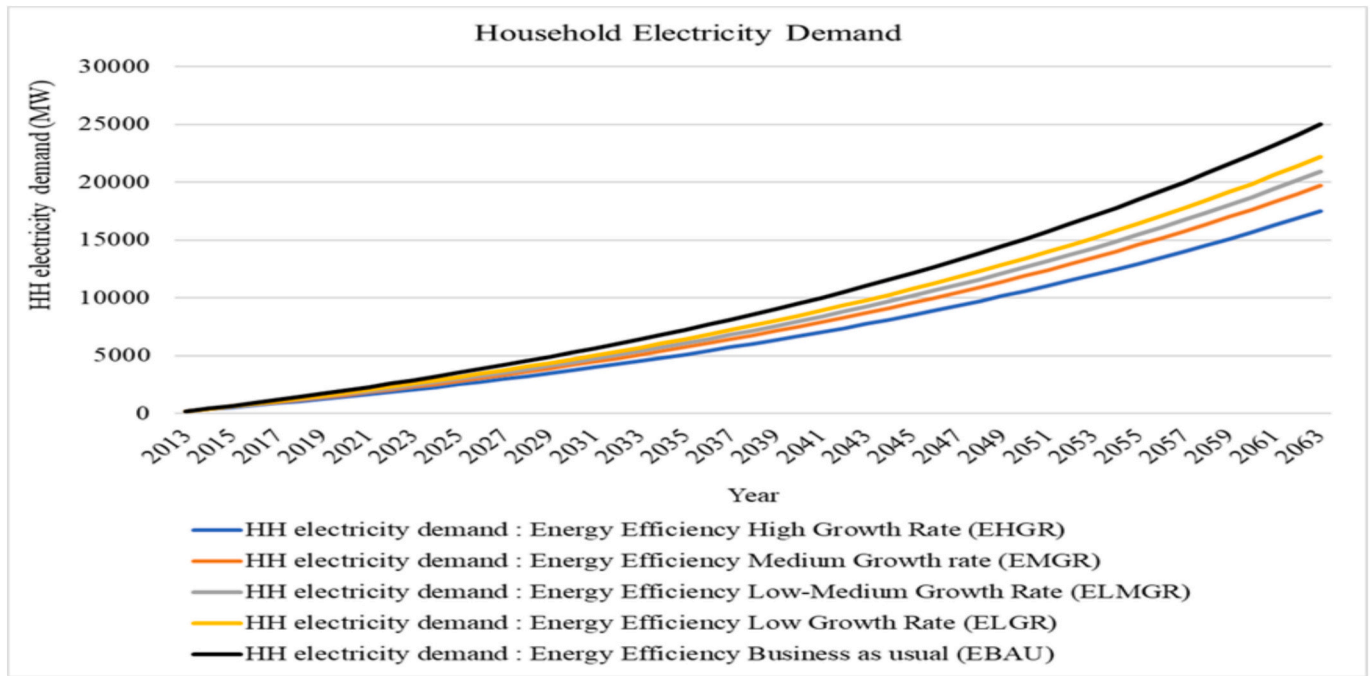


Fig. 7. Household electricity demand varied by demand efficiency factor.

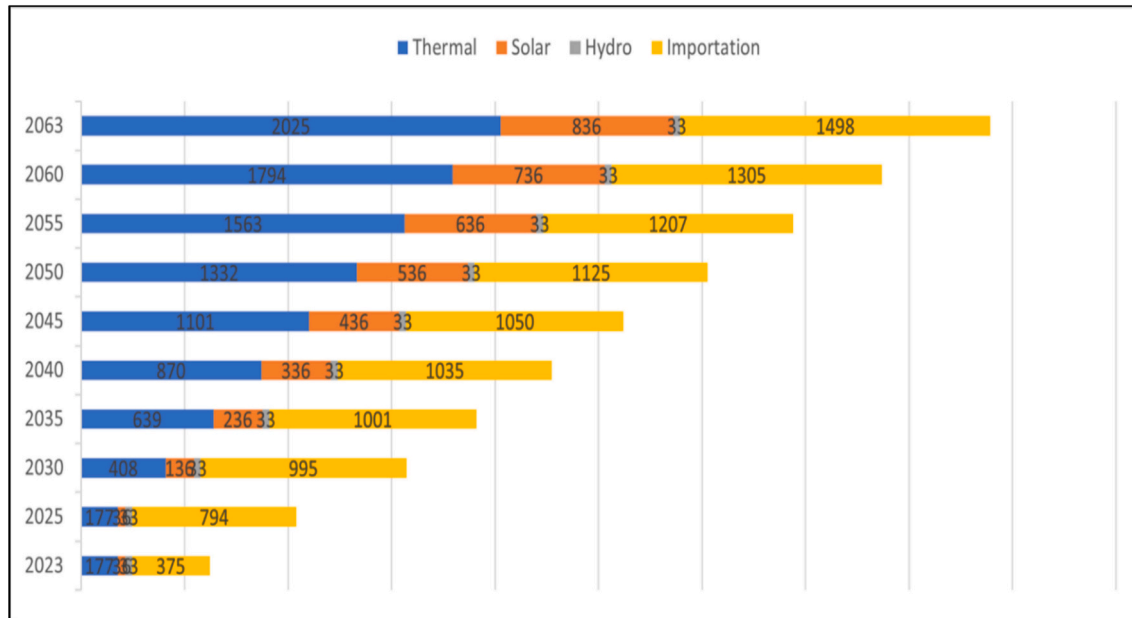


Fig. 8. Energy generation mix plan under business-as-usual scenario.

demand and affect them differently. Furthermore, evidence from [Li et al. \(2025\)](#) revealed that optimizing the electrical grid through densification and reduction of electric power transmission and distribution losses is a great gain to energy sustainability. Similarly, [Akpahou et al. \(2024\)](#) reported that efforts to reduce transmission losses will not only reduce dependence on electricity imports but also significantly increase domestic production, thereby ensuring the resilience and stability of Benin's energy system. Generally, a few studies have only applied the SD methodology to investigate the sustainability of energy security in the SSA region, especially in West Africa. Thus, the current study has filled a significant gap in providing a robust framework-SD - toward sustainable energy transition in Benin.

Limitation

The study has a caveat on the availability of complete energy and socio-techno-economic data. There is a dearth of energy data, especially on the rates of change in household electricity demand, supply rates, and consumption in Benin. However, data from 2013 are relatively stable, and some generation and supply data were found to be complete; hence, 2013 was employed as the base year. Also, where there exist data gaps, informed assumptions based on literature were made, recognizing that these gaps might influence the model's long-term behaviour, particularly for parameters such as electricity demand and supply. The choice of 2013 as the base year is based on the availability of relevant data for that

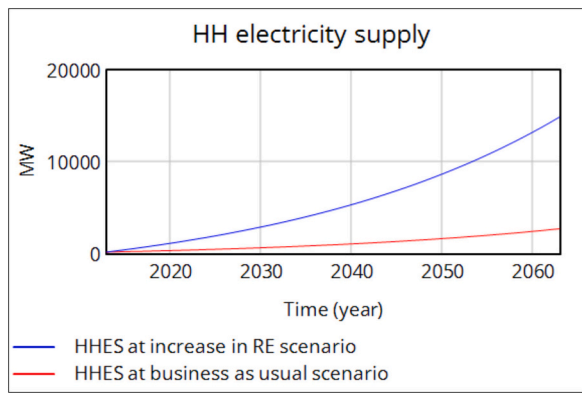


Fig. 9. Household Electricity Supply under Business-As-Usual and Increase in RE Scenario.

period. However, given ongoing socio-economic changes, data from 2013 were relatively stable, and some generation and supply data were found to be complete; hence, 2013 was employed as the base year. Another limitation is the long-term modelling framework, which could introduce some anomalies in the short-term. That is, the dynamics and changes in the short-term electricity planning in Benin could have been overlooked. Moreover, the SD model has been used only to explore the power sector, leaving out other energy sectors, such as clean cooking and transport. This was due to the limited time required to achieve the study's objectives and the availability of relevant data in the aforementioned sectors.

Conclusion

The focal point of this research was to develop a comprehensive SDs model that investigates the sustainable energy agenda of the Benin Republic in the household sector. The objectives of the study were to investigate the relevant socio-economic variables and how they influence the overall sustainable energy pathways through a system thinking

approach; to evaluate the household energy demand pattern and forecast in the medium- and long- term of the Benin energy sector through a SDs framework; and to develop a SDs model framework that explores the appropriate energy mix for achieving sustainable energy security in the household energy sector of Benin.

The simulations enabled the identification of several major trends and implications. Increasing population and economic growth are putting significant pressure on electricity demand. According to projections, the population of Benin will reach 15.98 million inhabitants in 2030 and 39.62 million in 2063, resulting in an estimated household

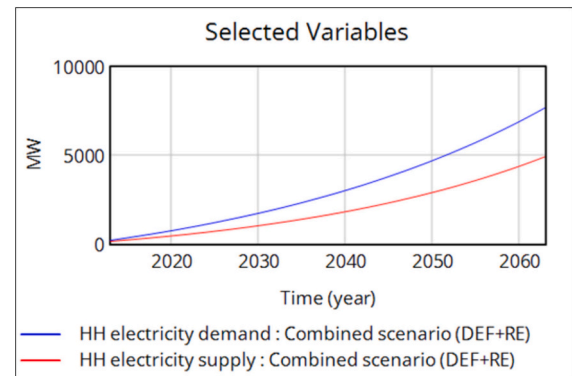


Fig. 11. HHED and HHES under combined scenario (DEF & RE).

Table 5

Values of transmission losses used for sensitivity analysis.

SN	Transmission losses	Annotation
1	0.215	Transmission losses Business as Usual (TBAU)
2	0.15	Transmission losses Low-Medium Decrease Rate (TLMDDR)
3	0.1	Transmission losses Low Decrease Rate (TLDR)

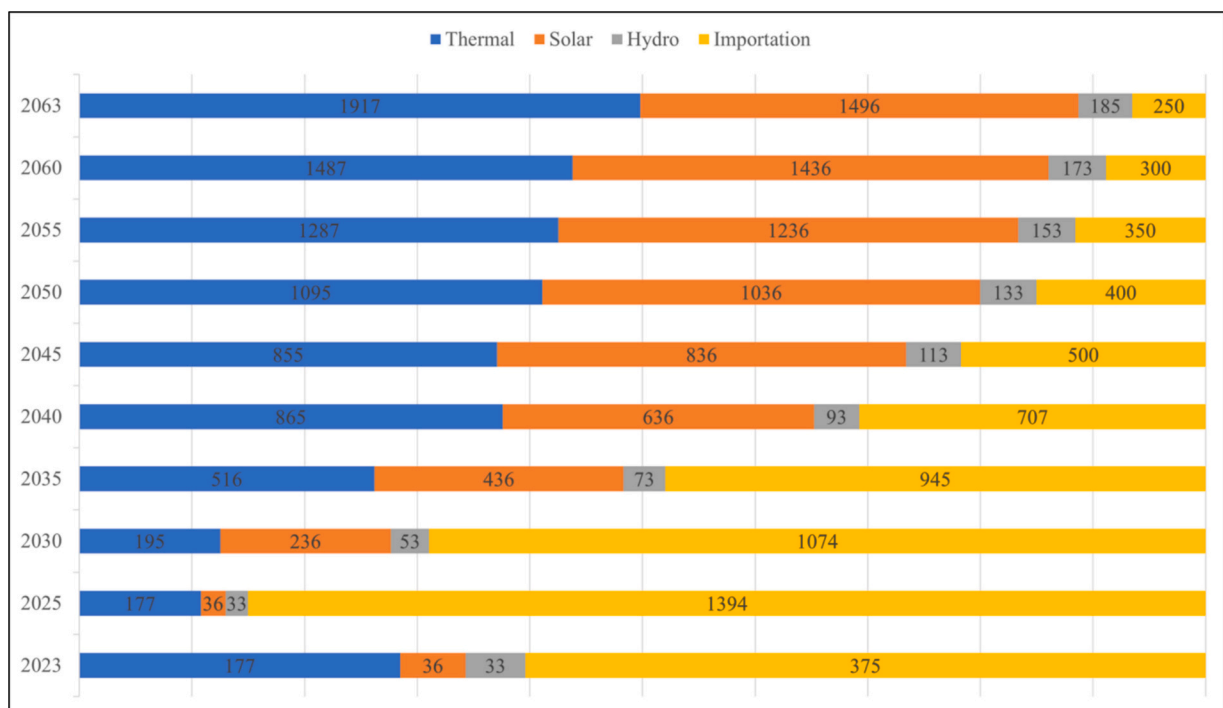


Fig. 10. Energy Mix Generation Plan for the Energy Security.

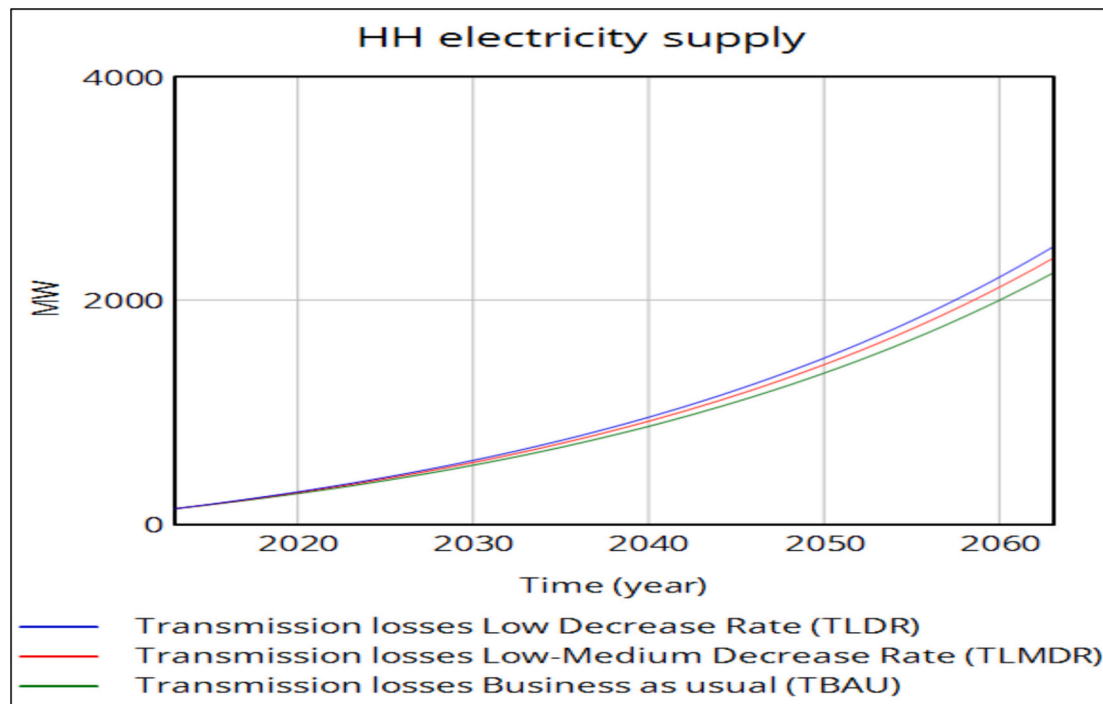


Fig. 12. Effect of transmission losses on electricity supply.

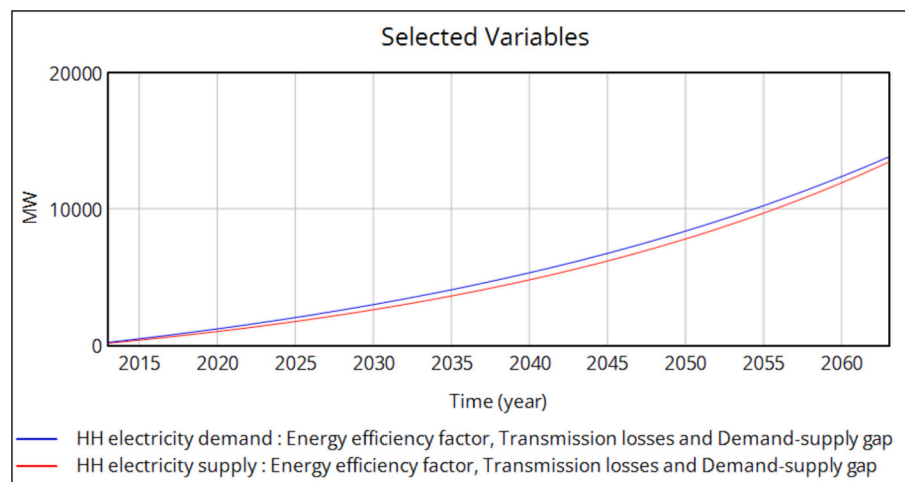


Fig. 13. Energy efficiency factor for supply and demand.

electricity demand of 5249.85 MW in 2030 and 24,991.6 MW in 2063 in a reference scenario. Improving energy efficiency helps slow demand growth by reducing energy intensity. An increase in the demand efficiency factors up to 26% could reduce demand by 7497.8 MW in 2063. Transmission losses (TL) significantly reduce electricity supply; reducing transmission losses to 15% and 10% (currently estimated at 21.5% of electric outputs in 2019, well above the African average of 8.3%) would improve energy availability from 523.49 MW to 547.83 MW and from 2238.85 MW to 2370.5 MW with TL at 15%, by 2030 and 2063, respectively, and to 566.55 MW and 2471.76 MW with TL at 10%, by 2030 and 2063 in that order, without massive investments in production. The energy mix must evolve toward a greater share of renewable energies. Analysis of the current scenario shows a high reliance on imports, with 72.11% of electricity produced from natural gas, followed by diesel and fuel oil, and only 4.55% from renewables in 2022. If current trends continue, the goal of universal access to electricity by 2035 seems unrealistic, as reliance on imports would persist in 2063

without accelerated investment in renewable energy. Simulations indicate that if the share of renewable energy reaches 35% in 2030 and continues to grow to 100% in 2063, dependence on imports could be reduced by up to 250 MW. However, thermal capacity will remain essential in the short and medium term to guarantee the base load of the national electricity grid. Improving the demand efficiency factor to 30% and increasing renewable energy to 45.40% in the energy mix would reduce the demand-supply gap to around 698 MW in 2063. Additionally, reducing transmission losses to 10%, combined with these measures, would shrink the gap to only 51 MW in 2063.

For future research, we recommend conducting an SDs study that captures the total electricity demand while illustrating the specific contributions and influences of each energy source of fossil fuels (natural gas and oil) and renewables (solar, wind, biomass, and small hydro). This approach would enable a more detailed understanding of how each resource behaves within the overall energy mix. Additionally, we suggest expanding the model boundaries. Given the evolving nature

of the energy system, it will be essential to integrate a broader range of policies over time. Expanding the model to include other energy sources such as LPG, charcoal, wood energy, and electricity exports, along with socio-economic and technical parameters like electricity prices and generation costs, would provide valuable insights and support the development of more comprehensive and responsive energy policies.

Policy recommendations

To ensure sustainable energy security, these recommendations are made:

- Accelerate investments in renewable energy, including solar photovoltaic and small and medium-sized hydroelectric power.
- Strengthen energy efficiency through incentive policies and increased consumer awareness.
- Reduce transmission losses by optimizing and modernizing the electricity network.
- Diversify energy sources to limit dependence on imports and ensure affordable and stable electricity.
- Use System Dynamics models to inform energy planning and improve public policy decision-making.

CRediT authorship contribution statement

Marius Esdras Tokpassi: Writing – original draft, Visualization, Validation, Formal analysis. **Babajide Epe Shari:** Writing – review & editing, Visualization, Supervision, Data curation. **Olayinka S. Ohunakin:** Writing – review & editing, Visualization, Supervision, Data curation, Conceptualization. **Kokou Amega:** Writing – original draft, Data curation. **Yendoubé Lare:** Writing – review & editing, Data curation. **Abdellatif Zerga:** Writing – review & editing, Visualization. **Abdellah Benyoucef:** Writing – review & editing, Visualization, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no conflicting financial interests that could have appeared to hamper the work reported in this paper.

Data availability

All data utilized for this work are available upon request.

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